

AI ASSISTANT CODING

ASSIGNMENT-16

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Batch:22

Task Description-1: (Design Database

Schema for Student Management

System) Instructions:

- Using AI tools, generate CREATE TABLE statements for a Student Management Database.
- Tables: Students, Courses, Enrollments.
- Include primary keys, foreign keys, and suitable data types.

Ensure the schema follows normalization rules (1NF, 2NF, 3NF).

Code:

The screenshot displays an AI-assisted coding interface. On the left, a file explorer shows a file named 'assl 15.py'. The main editor window, titled 'SQLite Untitled-1', contains the following SQL code:

```
1 -- SQLite
2 --Task Description-1: (Design Database Schema for Student Management
3 --Instructions:
4 --Using AI tools, generate CREATE TABLE statements for a Student
5 --Tables: Students, Courses, Enrollments.
6 --Include primary keys, foreign keys, and suitable data types.
7 --Ensure the schema follows normalization rules (1NF, 2NF, 3NF).
8 CREATE TABLE Students (
9     student_id INTEGER PRIMARY KEY AUTOINCREMENT,
10    first_name TEXT NOT NULL,
11    last_name TEXT NOT NULL,
12    email TEXT UNIQUE NOT NULL,
13    date_of_birth DATE NOT NULL
14 );
15 CREATE TABLE Courses (
16     course_id INTEGER PRIMARY KEY AUTOINCREMENT,
17     course_name TEXT NOT NULL,
18     course_code TEXT UNIQUE NOT NULL,
19     credits INTEGER NOT NULL
20 );
21 CREATE TABLE Enrollments (
22     enrollment_id INTEGER PRIMARY KEY AUTOINCREMENT,
23     student_id INTEGER NOT NULL,
24     course_id INTEGER NOT NULL,
25     enrollment_date DATE NOT NULL,
26     FOREIGN KEY (student_id) REFERENCES Students(student_id),
27     FOREIGN KEY (course_id) REFERENCES Courses(course_id)
28 );
29
```

On the right, the 'SQL' pane shows the generated SQL statements for each table, with a progress indicator '1 - 0 of 0'. The bottom status bar indicates 'Ln 14, Col 3 Spaces: 4 UTF-8 LF SQLite memory'.

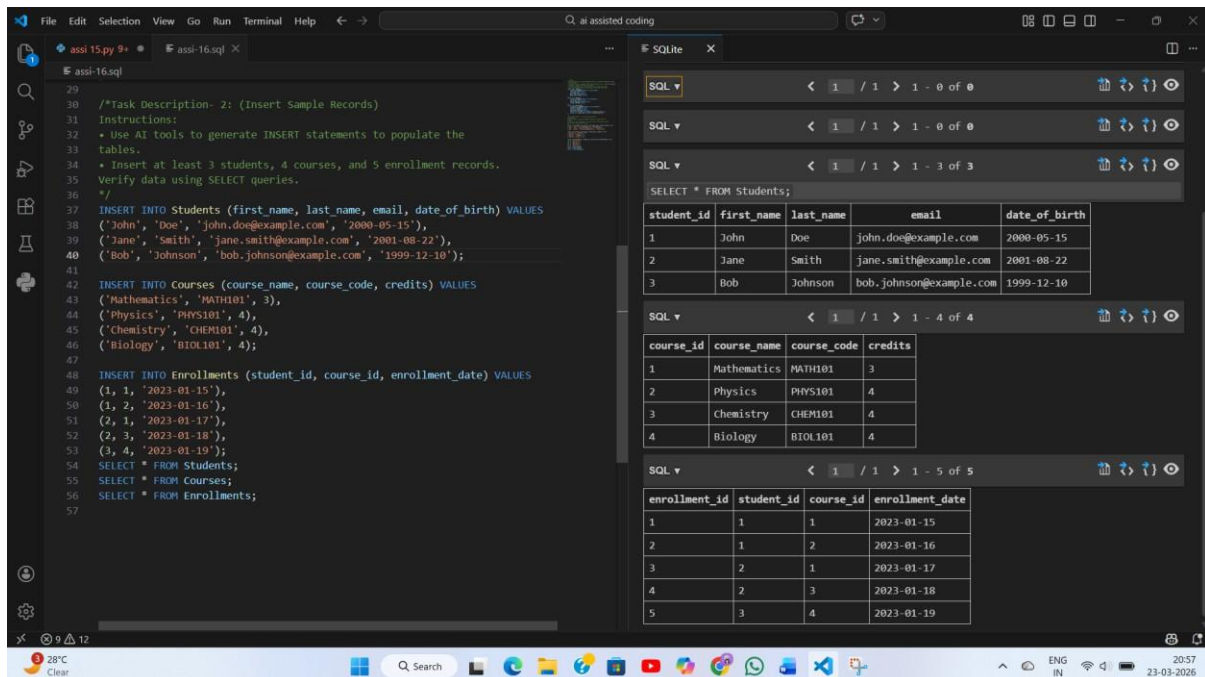
Task Description- 2: (Insert Sample Records)

Instructions:

- Use AI tools to generate INSERT statements to populate the tables.
- Insert at least 3 students, 4 courses, and 5 enrollment records.

Verify data using SELECT queries.

Code:



The screenshot shows a code editor with a file named 'assi-16.sql'. The code contains SQL statements to create a database schema and insert sample data. The schema includes tables for Students, Courses, and Enrollments. The code also includes instructions to use AI tools to generate INSERT statements and to verify data using SELECT queries. The right-hand pane shows the results of three SQL queries executed in the SQLite database. The first query shows the contents of the 'Students' table, the second shows the contents of the 'Courses' table, and the third shows the contents of the 'Enrollments' table.

```
29
30 /*Task Description- 2: (Insert Sample Records)
31 Instructions:
32 • Use AI tools to generate INSERT statements to populate the
33 tables.
34 • Insert at least 3 students, 4 courses, and 5 enrollment records.
35 Verify data using SELECT queries.
36 */
37 INSERT INTO Students (first_name, last_name, email, date_of_birth) VALUES
38 ('John', 'Doe', 'john.doe@example.com', '2000-05-15'),
39 ('Jane', 'Smith', 'jane.smith@example.com', '2001-08-22'),
40 ('Bob', 'Johnson', 'bob.johnson@example.com', '1999-12-10');
41
42 INSERT INTO Courses (course_name, course_code, credits) VALUES
43 ('Mathematics', 'MATH101', 3),
44 ('Physics', 'PHYS101', 4),
45 ('Chemistry', 'CHEM101', 4),
46 ('Biology', 'BIOL101', 4);
47
48 INSERT INTO Enrollments (student_id, course_id, enrollment_date) VALUES
49 (1, 1, '2023-01-15'),
50 (1, 2, '2023-01-16'),
51 (2, 1, '2023-01-17'),
52 (2, 3, '2023-01-18'),
53 (3, 4, '2023-01-19');
54 SELECT * FROM Students;
55 SELECT * FROM Courses;
56 SELECT * FROM Enrollments;
57
```

SQL query results:

Query 1: SELECT * FROM Students;

student_id	first_name	last_name	email	date_of_birth
1	John	Doe	john.doe@example.com	2000-05-15
2	Jane	Smith	jane.smith@example.com	2001-08-22
3	Bob	Johnson	bob.johnson@example.com	1999-12-10

Query 2: SELECT * FROM Courses;

course_id	course_name	course_code	credits
1	Mathematics	MATH101	3
2	Physics	PHYS101	4
3	Chemistry	CHEM101	4
4	Biology	BIOL101	4

Query 3: SELECT * FROM Enrollments;

enrollment_id	student_id	course_id	enrollment_date
1	1	1	2023-01-15
2	1	2	2023-01-16
3	2	1	2023-01-17
4	2	3	2023-01-18
5	3	4	2023-01-19

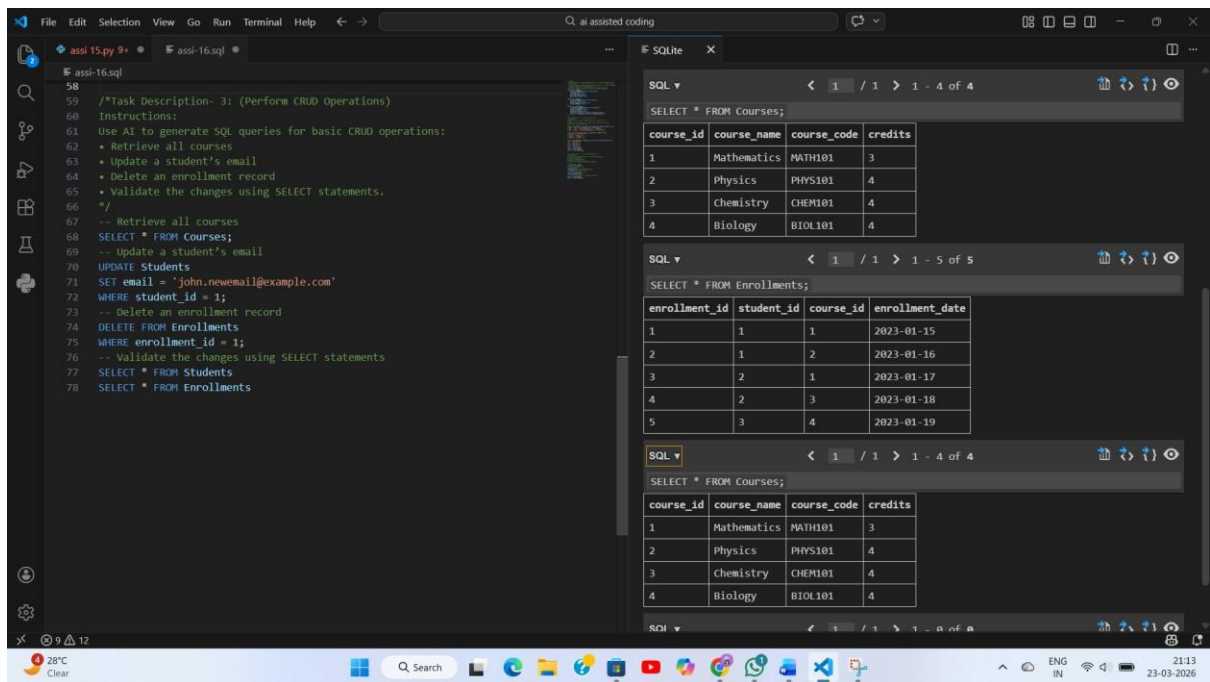
Task Description- 3: (Perform CRUD Operations)

Instructions:

Use AI to generate SQL queries for basic CRUD operations:

- Retrieve all courses
- Update a student's email
- Delete an enrollment record
- Validate the changes using SELECT statements.

Code:



Task Description- 4: (Execute Aggregate Queries)

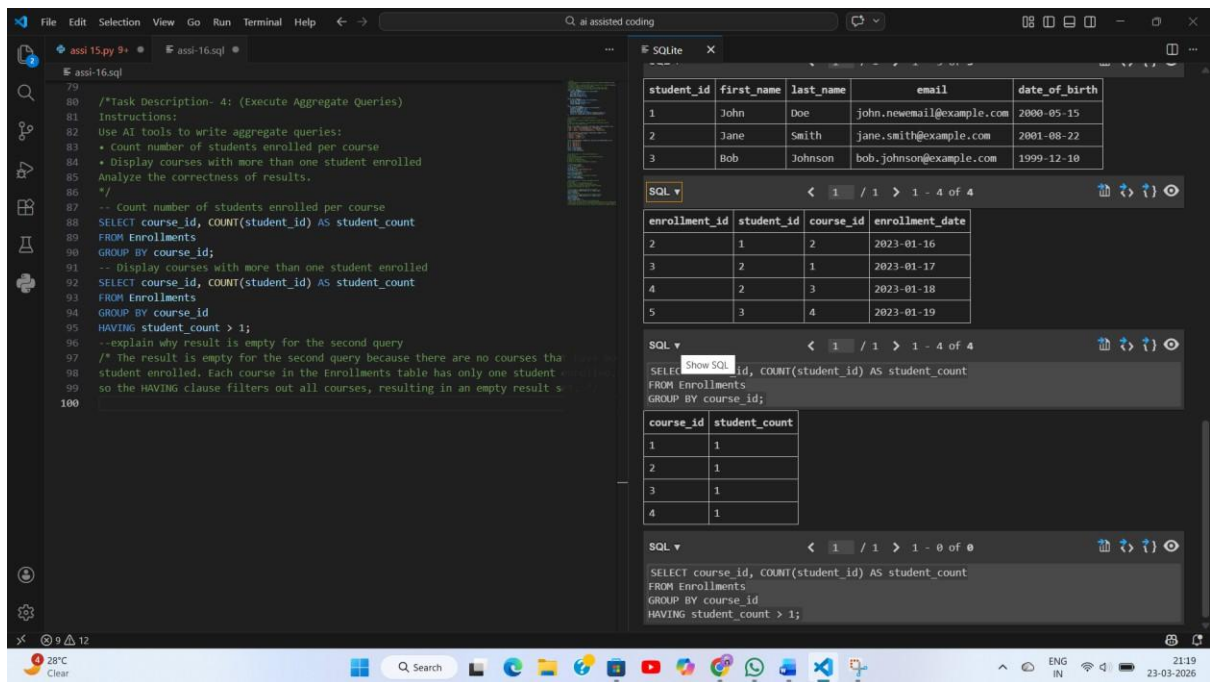
Instructions:

Use AI tools to write aggregate queries:

- Count number of students enrolled per course
- Display courses with more than one student enrolled

Analyze the correctness of results.

Code:



Task Description- 5: (Query Optimization)

Instructions:

- Use AI tools to analyze query performance.
- Rewrite inefficient sub queries using JOINS and test both queries for correctness.

Code:

File Edit Selection View Go Run Terminal Help ← → Q ai assisted coding

assi-15.py 9+ • assi-16.sql •

```
100
101
102 /*Task Description- 5: (Query Optimization)
103 Instructions:
104 • Use AI tools to analyze query performance.
105 • Rewrite inefficient sub queries using JOINS and test both queries
106 for correctness.*/
107 -- Inefficient subquery using IN
108 SELECT course_name
109 FROM Courses
110 WHERE course_id IN (SELECT course_id FROM Enrollments WHERE student_id = 2);
111 -- Optimized query using JOIN
112 SELECT c.course_name
113 FROM Courses c
114 JOIN Enrollments e ON c.course_id = e.course_id
115 WHERE e.student_id = 1;
116
```

SQLite X

3	2	1	2023-01-17
4	2	3	2023-01-18
5	3	4	2023-01-19

SQL ▾ < 1 / 1 > 1 - 4 of 4

course_id	student_count
1	1
2	1
3	1
4	1

SQL ▾ < 1 / 1 > 1 - 0 of 0

SQL ▾ < 1 / 1 > 1 - 2 of 2

```
SELECT course_name
FROM Courses
WHERE course_id IN (SELECT course_id FROM Enrollments WHERE student_id = 2);
```

course_name
Mathematics
Chemistry

SQL ▾ < 1 / 1 > 1 - 1 of 1

```
SELECT c.course_name
FROM Courses c
JOIN Enrollments e ON c.course_id = e.course_id
WHERE e.student_id = 1;
```

course_name
Physics

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